

Bay En Wang

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GitHub: <https://github.com/bayen0102/mywebsite/> | Portfolio: <https://mywebsite-s9jv.onrender.com>

Summary

Computer Science graduate and M.S. Informatics student with experience building real-time communication systems, data pipelines, and backend applications using Python, SQL, and Flask.

Education

San Jose State University

Master's in Informatics

Sep 2026 - May 2028(Expected)

University of California, Santa Cruz

Bachelor of Science in Computer Science

GPA: 3.31/4.00

Dean's Honors (Spring 2026)

Sep 2024 - Jun 2026

Fullerton College

Associate of Science in Computer Science and Mathematics (Multiple Degree)

GPA: 3.56 / 4.00

Aug 2019 - May 2024

Skills

Programming: Python, C++, SQL

Backend: FastAPI, Flask, REST APIs

Databases & Caching: PostgreSQL, MySQL, SQLite, Redis

Data: ETL, Data Pipelines, Data Analysis

Networking: TCP/IP, Socket Programming

Tools: Git, GitHub, Linux, Docker, Docker Compose, GitHub Actions, pytest

Projects

Real-Time Communication and Data Pipeline System

Jan 2026 - Present

- Engineered a hybrid messaging system using **Python/TCP**, combining centralized peer discovery with persistent peer-to-peer connections for real-time communication
- Migrated persistence from **SQLite to PostgreSQL**, integrated **Redis** for user presence, and developed **FastAPI REST APIs** for message history and analytics
- Containerized the TCP server, API, PostgreSQL, and Redis using **Docker Compose** and added structured JSON logging for backend observability
- Built automated tests with **pytest** and a **GitHub Actions CI pipeline** that provisions PostgreSQL/Redis and validates the backend on every push and pull request

Personal Portfolio Website

- Designed and developed a personal portfolio website using **Flask, HTML, CSS** to showcase technical projects and system designs
- Integrated project pages highlighting real-time communication systems and data pipeline architecture
- Structured content to present technical skills, including **networking (TCP)**, **SQL analytics**, and **system design**
- Deployed the application on Render for public access and demonstration

Experience & Leadership

Magikid Robotics Lab - Tutor

- Taught students **Python, Scratch**, and basic robotics assembly, building foundational programming and engineering concepts
- Guided students in understanding algorithmic logic and debugging in Python, helping them develop independent problem-solving skills
- Adapted teaching strategies to different learning styles, improving student's logical thinking and overall learning outcomes

Chinese Student Association — President (Fullerton college)

- Led a student organization by coordinating events, managing communication, and overseeing team operations
- Organized and executed campus events, increasing student participation and engagement
- Collaborated with team members to plan and deliver structured activities, demonstrating leadership, teamwork, and organizational skills

Teachamp Tea shop — Manager

- Analyzed sales and customer behavior data to identify high-demand products
- Optimized inventory decisions and staffing during peak hours
- Improved operational efficiency through data-driven decision making